

Fundamental theorem of calculus:

①

Assume $f(x)$ is a continuous function over $[a, b]$, F is an antiderivative of f if and only if

$$\int_a^b f(x) dx = F(b) - F(a)$$

1. Tabel of basic integrals that you should remember.

$$1. \int \# dx = \# x + C$$

$$2. \int u du = \frac{u^n}{n+1} + C$$

$$3. \int e^u du = e^u + C$$

$$4. \int \frac{du}{u} = \ln|u| + C$$

$$5. \int \sin u du = -\cos u + C$$

$$6. \int \cos u du = \sin u + C$$

$$7. \int \sec^2 u du = \tan u + C$$

$$8. \int \csc^2 u du = \cot u + C$$

2. Tabel of frequently used trig identities.

$$\sin^2 x + \cos^2 x = 1$$

$$\sin 2x = 2 \sin x \cos x$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

e.g solve the following integrals.

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$$1. \int (x^3 - 3x + 2 - \frac{2}{x}) dx$$
$$= \frac{x^4}{4} - \frac{3}{2}x^2 + 2x - 2\ln|x| + C$$

$$2. \int \frac{1}{x^2} e^{\frac{1}{x}} dx$$

we solve the integral with method of substitution.

$$\text{Let } \frac{1}{x} = u \Rightarrow -\frac{dx}{x^2} = du \Rightarrow \frac{dx}{x^2} = (-du)$$

$$\int e^{\frac{1}{x}} \left(\frac{dx}{x^2}\right) = -\int e^u du = -e^u + C$$
$$= -e^{\frac{1}{x}} + C$$

$$3. \int_{-1}^{-2} \left(\frac{5}{3} x^{\frac{2}{3}} - \frac{4}{x^3}\right) dx$$

$$= \int_{-1}^{-2} \left(\frac{5}{3} x^{\frac{2}{3}} - 4x^{-3}\right) dx = \left(\frac{5}{3} \cdot \frac{3}{5} x^{\frac{5}{3}} + \frac{4}{2} x^{-2}\right) \Big|_{-1}^{-2}$$

$$= \left(x^{\frac{5}{3}} + \frac{2}{x^2}\right) \Big|_{-1}^{-2} = -\sqrt[3]{4} + \frac{1}{2} - (-1 + 2) = -\left(\sqrt[3]{4} + \frac{1}{2}\right)$$

$$4. \int_0^{\infty} \lambda e^{-\lambda x} dx \underset{\uparrow}{=} \lim_{t \rightarrow \infty} \int_0^t \lambda e^{-\lambda x} dx$$

This is an improper integral

for this question we will assume $\lambda > 0$.

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$$\lim_{t \rightarrow \infty} \int_0^t \lambda e^{-\lambda x} dx = \lim_{t \rightarrow \infty} -e^{-\lambda x} \Big|_0^t$$

$$= \lim_{t \rightarrow \infty} -e^{-\lambda t} + e^0 = 1$$

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Integration By parts formula

$$\int u dv = uv - \int v du$$

eg. $\int x \cos x dx$

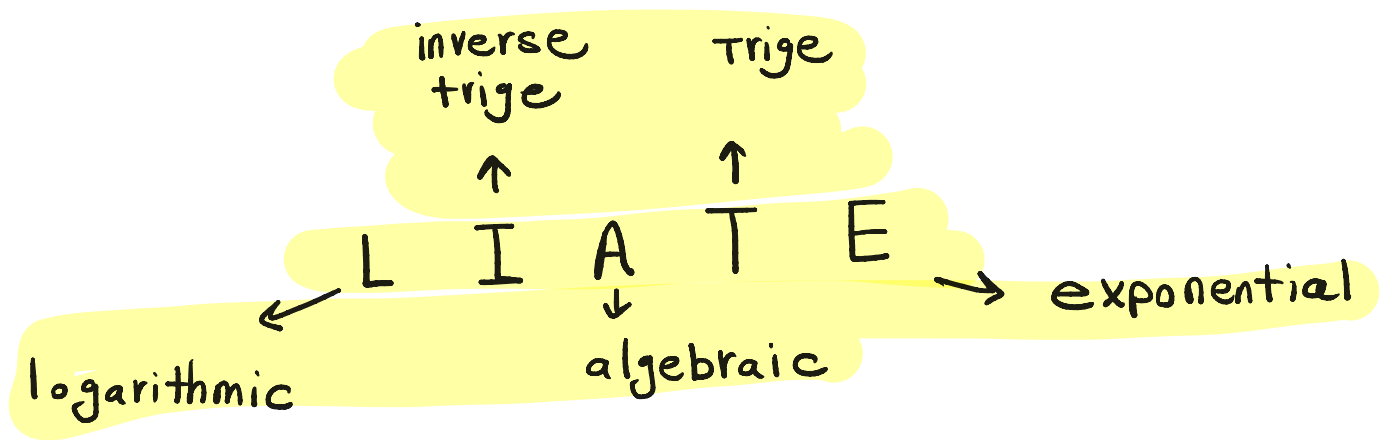
$$u = x \quad dv = \cos x dx$$

$$du = dx \quad v = \sin x$$

$$\begin{aligned} \int x \cos x &= x \sin x - \int \sin x dx \\ &= x \sin x + \cos x + C \end{aligned}$$

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note: A mnemonic that helps to understand in integration by parts what to choose as u is **LIATE**. whatever comes first we choose that one as u [sth that we should take derivative of]



For example $\int x \ln x \, dx$

then $\ln x$ is logarithmic and x is algebraic,
hence we should choose $\ln x$ as u , $x \, dx$ as dv .

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table rule: we create a table with three columns, first column is only sign which starts by a positive and second one is negative and then it alternates.

In the D column on the first row write what you are taking

S	D	∫
+	→ u	dv
-	→ du	v
+	→ d ² u	∫ v
-	→ d ³ u	∫ ∫ v
+	→ ⋮	⋮
⋮	→ 0	⋮

for u, and take its derivative

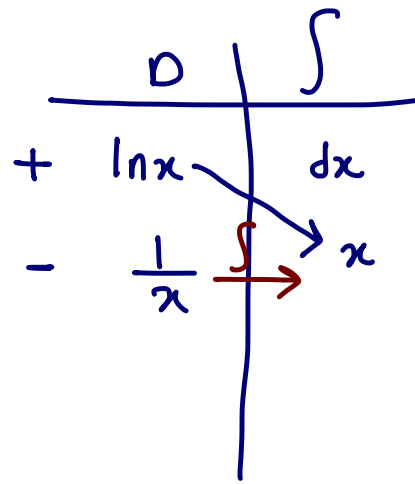
and write it on second row, keep doing it, till derivative reaches zero [If that is possible]

In the third column write dv on the first row, take its integral and write the result in the second row, take integral of the second row and write in the third row and continue.

$$\int u dv = uv - \int v du + \int \int v d^2 u \dots$$

note: If the function we choose as u is a function of $\ln(x)$, then we only take derivative once.

eg. $\int \ln x \, dx$



$$\int \ln x \, dx = x \ln x - \int dx$$

$$= x \ln x - x + C$$

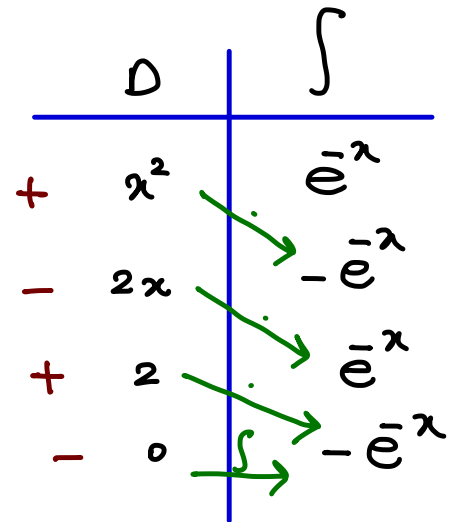
eg. $\int_0^1 x^2 e^{-x} \, dx$

$$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \Big|_0^1$$

$$= \frac{-1}{e^x} [x^2 + 2x + 2]_0^1$$

$$= -\frac{1}{e} [1 + 2 + 2] + [2]$$

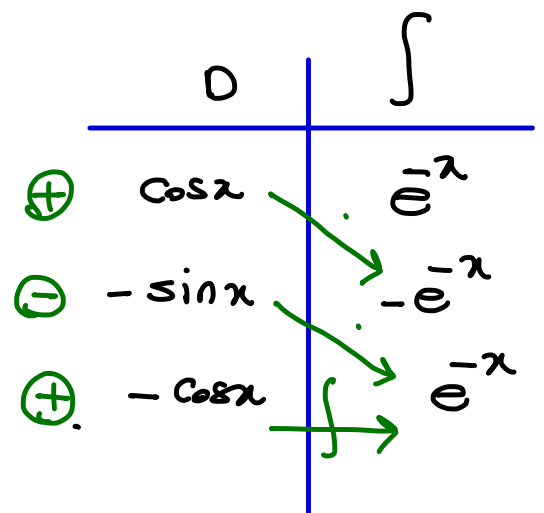
$$= 2 - \frac{5}{e}$$



eg. $\int \frac{\cos x}{e^x} \, dx$

$$\int e^{-x} \cos x \, dx = -\cos x e^{-x}$$

$$+ \sin x e^{-x} - \int \cos x e^{-x} \, dx$$



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$$2 \int e^{-x} \cos x \, dx = e^{-x} [\sin x - \cos x]$$

$$\int e^{-x} \cos x \, dx = \frac{e^{-x}}{2} [\sin x - \cos x] + C.$$

eg. $\int x^2 \ln x \, dx$

$$= \frac{x^3}{3} \ln x - \int \frac{x^2}{3} \, dx$$

D	$\int$
$\oplus \ln x$	$x^2 \, dx$
$\ominus \frac{1}{x}$	$\frac{x^3}{3}$

Diagram illustrating integration by parts for $\int x^2 \ln x \, dx$. The table shows the derivative (D) and integral ($\int$) of the functions. An arrow points from $\ln x$ to $\frac{x^3}{3}$, and another arrow points from $\frac{1}{x}$ to $\frac{x^3}{3}$.

$$\int x^2 \ln x \, dx = \frac{x^3}{3} \ln x - \frac{x^3}{9} + C$$

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eg. $\int (\ln x)^2 dx$

	D	$\int$
$\oplus$	$(\ln x)^2$	dx
$\ominus$	$\frac{2 \ln x}{x}$	x

$$\int (\ln x)^2 dx = x(\ln x)^2 - 2 \int \ln x dx$$

$$\int \ln x dx = x \ln x - \int dx$$

$$\int \ln x dx = x \ln x - x + C$$

	D	$\int$
$\oplus$	$\ln x$	dx
$\ominus$	$\frac{1}{x}$	x

$$\int (\ln x)^2 dx = x(\ln x)^2 - 2x \ln x + 2x + C$$

Review on the Geometric series

⑨

A Geometric series is of the form $\sum_{n=m}^{\infty} q^n$.

If $|q| < 1$, then the series is convergent, and

$$\sum_{n=m}^{\infty} q^n = \frac{q^m}{1-q}$$

Note: In the above series q is the constant ratio of the geometric sequence $a_n = q^n$; $n=m, \dots$

$$q^m, q^{m+1}, q^{m+2}, \dots$$

That means every term after the first term is obtained by multiplying the previous term by q .

In fact, if $\sum_{n=m}^{\infty} a_n$ is a geometric

series, if $\left| \frac{a_{n+1}}{a_n} \right| < 1$, then it's convergent

Let $q = \frac{a_{n+1}}{a_n}$, then $\sum_{n=m}^{\infty} a_n = \frac{q^m}{1-q}$

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So for solving a geometric series

always first find ratio $\frac{a_{n+1}}{a_n}$. For

example for series $\sum_{k=0}^{\infty} q^{3k}$

$$\frac{a_{k+1}}{a_k} = \frac{q^{3k+3}}{q^{3k}} = q^3, \text{ hence}$$

$$\sum_{k=0}^{\infty} q^{3k} = \frac{(q^3)^0}{1 - q^3} = \frac{1}{1 - q^3}$$

Eg. In the following examples assume

②

$$0 < p < 1.$$

$$(a) \sum_{k=0}^{\infty} (1-p)^{2k}$$

Since $0 < p < 1$, $0 < 1-p < 1$. Therefore the above series is convergent.

$$\frac{a_{k+1}}{a_k} = \frac{(1-p)^{2(k+1)}}{(1-p)^{2k}} = \frac{(1-p)^{2k+2}}{(1-p)^{2k}} = (1-p)^2$$

$$\sum_{k=0}^{\infty} (1-p)^{2k} = \frac{\left((1-p)^2\right)^0}{1 - (1-p)^2} = \frac{1}{1 - (1-p)^2}$$

$$(b) \sum_{n=1}^{\infty} n(1-p)^{n-1}$$

Note that $n(1-x)^{n-1} = \frac{d}{dx} [(1-x)^n]$

$$\begin{aligned} \sum_{n=1}^{\infty} n(1-p)^{n-1} &= - \sum_{n=1}^{\infty} \frac{d}{dp} (1-p)^n \\ &= - \frac{d}{dp} \left(\sum_{n=1}^{\infty} (1-p)^n \right) \\ &= - \frac{d}{dp} \left(\frac{(1-p)}{1-(1-p)} \right) \\ &= - \frac{d}{dp} \left[\frac{1-p}{1-1+p} \right] \\ &= - \frac{d}{dp} \left[\frac{1-p}{p} \right] \\ &= - \left(\frac{-p - (1-p)}{p^2} \right) = \frac{1}{p^2} \end{aligned}$$

(c) $\sum_{k=2}^{\infty} \frac{\lambda^k}{k!}$ Assume $\lambda > 0$.

Recall $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

Hence $e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$

$$e^{\lambda} = \frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \sum_{k=2}^{\infty} \frac{\lambda^k}{k!}$$

$$e^{\lambda} = 1 + \lambda + \sum_{k=2}^{\infty} \frac{\lambda^k}{k!}$$

$$\Rightarrow e^{\lambda} - (1 + \lambda) = \sum_{k=2}^{\infty} \frac{\lambda^k}{k!}$$